58 c

UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2023/2024 FIRST SEMESTER EXAMINATION

COURSE: MTH316: INTRODUCTION TO MATHEMATICAL MODELLING
TIME ALLOWED: THREE HOURS (3 HRS)

Instructions: Answer question one (1) and any other three.

1. Consider a water mass, within which phytoplanktons grow and diffusion takes place. Assume that this water mass is surrounded by water in which plankton cannot survive. Part of the population is continuously lost to the surroundings owing to diffusion, the plankton patch would cease to exist unless reproduction within it counterbalances this loss. If the phytoplankton is limited to grow only in a one-dimensional region along the x-axis, and the region of bloom is limited to $(0, L)$, then the diffusion equation of the organism is given by:
$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} + \alpha C$; Where C is the concentration of the plankton, D is the diffusivity, α is the rate of growth and L is the size of the water mass.
(a) Find the solution of equation that vanishes at $x = 0$ and $x = L$ corresponding to the initial condition $C(x, 0) = f(x)$.
(b) What is the critical length, L_c, of the water mass in the region $0 \leq x \leq L$.

2. (a) What is mathematical modelling? Explain, with the aid of a flowchart, the three basic steps. 17.5
(b) Name and explain the two kinds of mathematical models

3. (a) List the four types of mathematical models in terms of classification. 8
(b) Consider whether the equation, $\frac{dN}{dt} = \pm\lambda N$, is linear or not. Solve the equation for both the positive and negative signs. Explain the nature of the solutions as $t \longrightarrow \infty$

4. (a) Consider whether the equation, $\frac{dQ}{dt} = KQ$, if K is the proportionality constant and Q is the amount present at time $t = 0$. Using the dimension analysis, show that K has a dimension of reciprocal time.
(b) A raindrop, beginning at rest, falls from a cloud 705.6m above the ground. Formulate the raindrop as a freely falling object, determine the time it will take to reach the ground. 8

5. Stokes's law states that for a spherical raindrops falling in still air and having diameter $D<0.762cm$, the acceleration due to gravity is opposed by an amount proportional to the velocity of the raindrop. Specifically, an amount equal to $\frac{0.329 \times 10^{-5}}{D^2}\frac{dx}{dt}$.
Thus $\frac{d^2x}{dt^2} = 980 - \frac{0.329 \times 10^{-5}}{D^2}\frac{dx}{dt}$
Solve the differential equation.
(b)Predict the terminal velocity,$(V_{term} = \frac{dx}{dt})$ i.e when $\frac{d^2x}{dt^2} = 0$, take $D = 0.5cm$

6. Use the Newton's second law of motion to formulate a model (write down the arising governing differential equations) to determine the period of oscillation and the tension in the string of a simple pendulum. Assume that the only forces acting on the mass are the tension, T in the string and the gravitational force, mg. The tension acts along the line of the string while the gravitational force acts vertically downward along the y-axis.

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